

Cloud LearnX

Mathematics

Similar Solids and Trigonometry — Practice Worksheet

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Topic: Similar Solids and Trigonometry

Instructions:

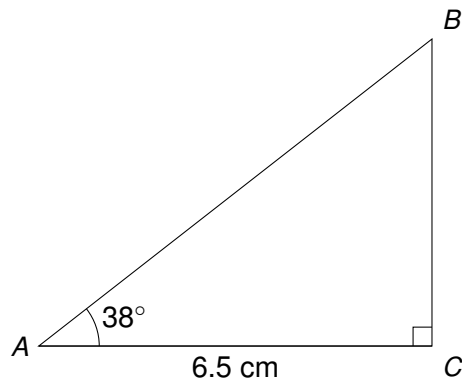
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** A conference organiser is setting up a projector screen stand for a Model United Nations debate. The support strut AB makes an angle of 38° with the horizontal base AC , as shown in the diagram.



$AC = 6.5$ cm and angle $BAC = 38^\circ$.

Work out the length of BC , giving your answer correct to 1 decimal place.

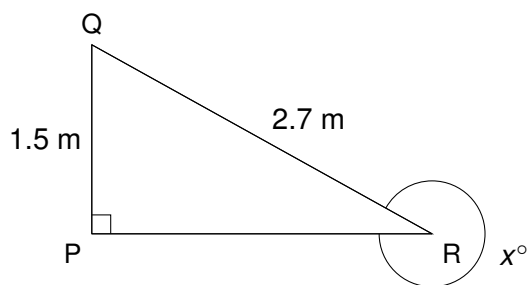
..... cm

(2)

(Total for Question 1 is 2 marks)

- 2** A badminton court has a straight net post PQ that stands vertically on the court, and a guy wire QR is attached from the top of the post to a point R on the ground, anchoring it in place.

$PQ = 1.5$ m and $QR = 2.7$ m. Triangle PQR has a right angle at P .



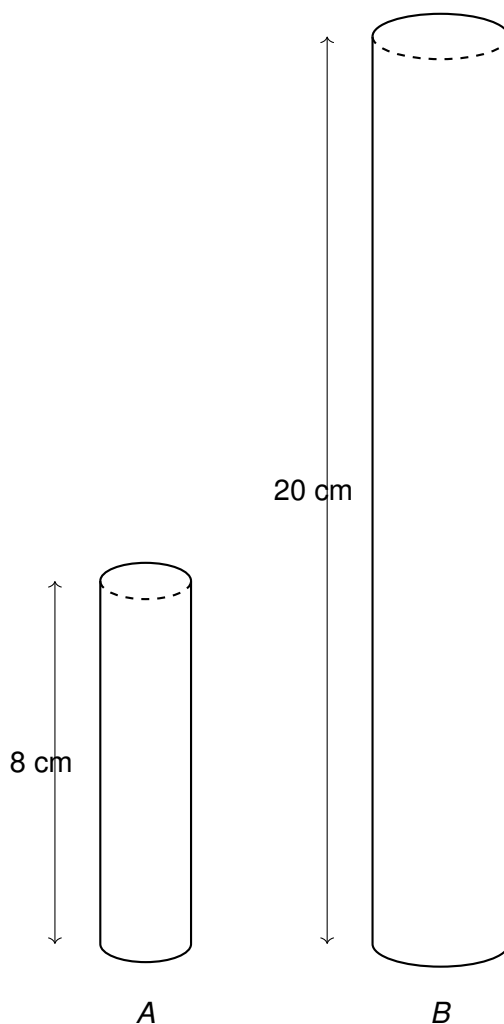
Work out the size of angle x , the angle between the guy wire QR and the ground.

$x = \dots\dots\dots^\circ$
(2)

(Total for Question 2 is 2 marks)

3 A science laboratory uses two similar cylindrical test tubes, made by the same manufacturer, for a chemistry experiment.

Test tube *A* has a length of 8 cm and test tube *B* is a larger, similar test tube with a length of 20 cm.



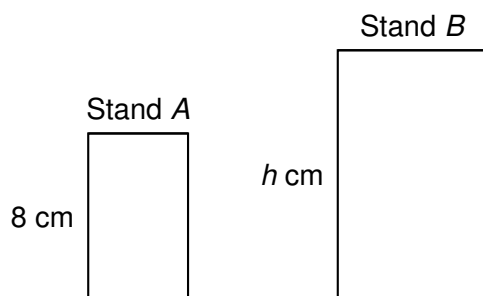
Find the linear scale factor from test tube *A* to test tube *B*.

.....

(2)

(Total for Question 3 is 2 marks)

- 4 A school science department has two similar equipment stands used to hold measuring cylinders during experiments. Stand *B* is an enlargement of Stand *A*.



The width of Stand *A* is 2.4 cm and the width of Stand *B* is 3.6 cm. The height of Stand *A* is 8 cm. Work out the height, *h*, of Stand *B*.

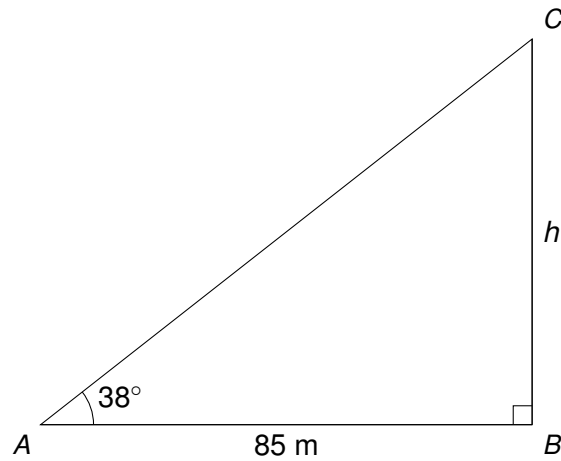
$$h = \dots\dots\dots \text{ cm}$$

(2)

(Total for Question 4 is 2 marks)

- 5** As part of a school science project, a team releases a weather balloon from a point B on level ground. An observer stands at point A , which is 85 m from the release point B , tracking the balloon's ascent for their Model United Nations climate-monitoring presentation.

When the balloon reaches point C directly above B , the angle of elevation of the balloon from the observer at A is 38° .



Work out the height, h , of the balloon above the ground at point C , giving your answer correct to 3 significant figures.

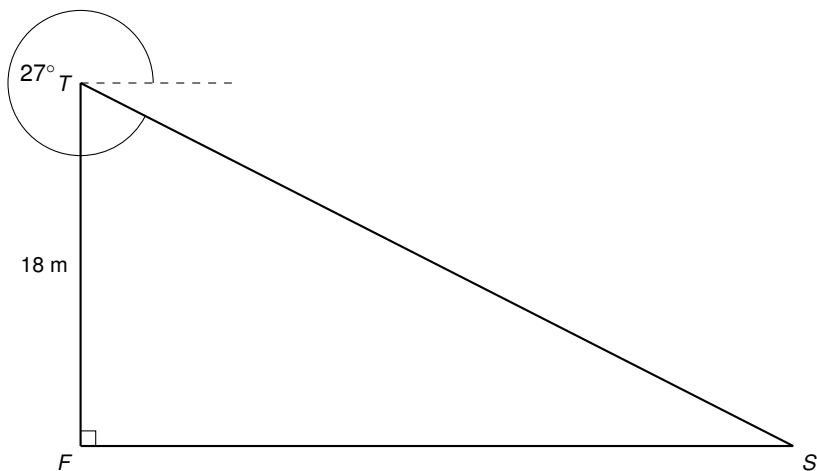
$$h = \dots\dots\dots \text{ m}$$

(3)

(Total for Question 5 is 3 marks)

- 6** A scientist is using a telescope mounted at the top of a laboratory tower to observe a specimen resting on a flat table in the courtyard below.

The telescope is at point T , which is 18 m above the level of the table. The specimen is at point S on the table. The angle of depression from T to S is 27° .



F is the point on the table directly below the telescope, so that TF is vertical and FS is horizontal.

Calculate the horizontal distance FS from the base of the tower to the specimen.

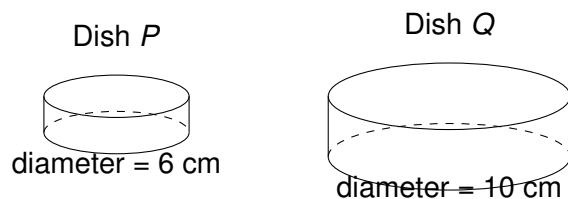
Give your answer correct to 3 significant figures.

..... m

(3)

(Total for Question 6 is 3 marks)

- 7 Two petri dishes used in a school science lab are mathematically similar circular cylinders, dish P and dish Q , as shown below.



Dish P has a diameter of 6 cm and a total surface area of 84.8 cm^2 .

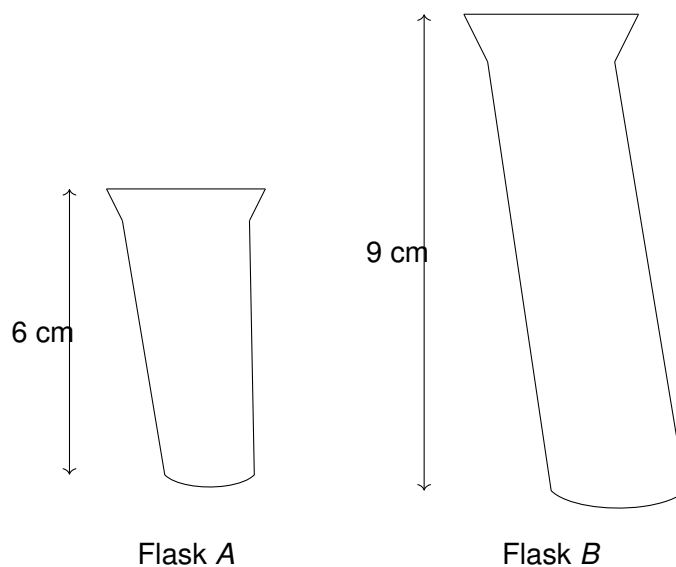
Dish Q is mathematically similar to dish P and has a diameter of 10 cm.

Work out the total surface area of dish Q .

..... cm^2
(3)

(Total for Question 7 is 3 marks)

- 8** Two laboratory conical flasks, *A* and *B*, are mathematically similar in shape.
 The height of flask *A* is 6 cm and the height of flask *B* is 9 cm.
 The volume of flask *A* is 80 cm³.

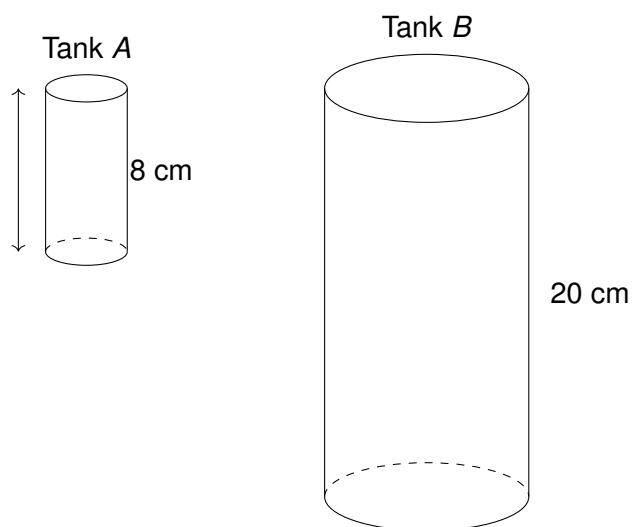


Calculate the volume of flask *B*.

..... cm³
 (3)

(Total for Question 8 is 3 marks)

- 9 Two cylindrical storage tanks used in a school science laboratory are mathematically similar. Tank *A* is a small prototype and Tank *B* is the full-size version used for storing samples.



Tank *A* has a height of 8 cm and a total surface area of 150 cm^2 .

Tank *B* has a height of 20 cm.

The volume of Tank *A* is 320 cm^3 .

- (a) Find the linear scale factor from Tank *A* to Tank *B*.

.....

(1)

- (b) Find the ratio of the surface area of Tank *A* to the surface area of Tank *B*, giving your answer in the form $1 : n$.

.....

(1)

- (c) Work out the volume of Tank *B*.

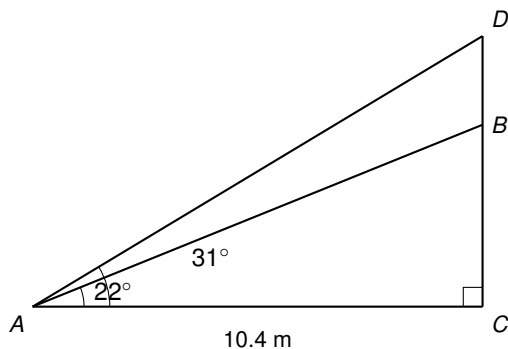
..... cm^3

(2)

(Total for Question 9 is 4 marks)

- 10** A scientist is setting up a ramp to move heavy equipment up onto a raised platform. The ramp is supported by a vertical strut BC , which stands on the horizontal ground at C and meets the platform at B . The foot of the ramp is at A , on the ground 10.4 m from the base of the strut, and the top of the ramp meets the platform at D , directly above C .

From A , the angle of elevation to the top of the strut B is 22° . From A , the angle of elevation to the top of the platform D is 31° .



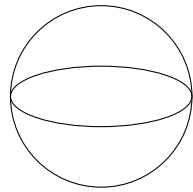
Work out the height of the platform, BD , giving your answer to 3 significant figures.

..... m

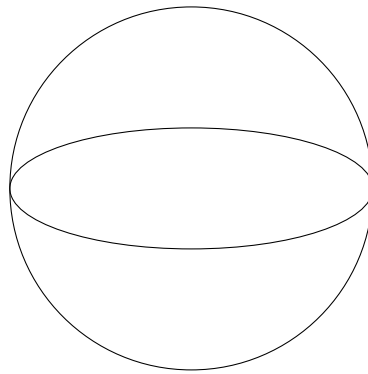
(4)

(Total for Question 10 is 4 marks)

- 11** For a Model United Nations science and culture exhibition, a student designs two similar spherical globe models to represent Earth at different scales for a UI/UX display stand.



Model A
Volume = 288 cm^3



Model B
Volume = 2304 cm^3

Model A and Model B are mathematically similar solids.

Model A has a volume of 288 cm^3 and Model B has a volume of 2304 cm^3 .

Find the linear scale factor from Model A to Model B.

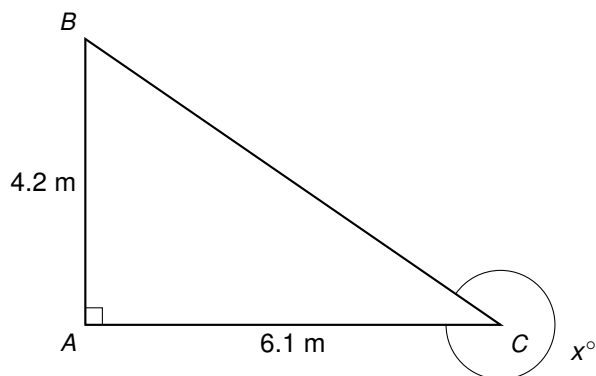
.....

(3)

(Total for Question 11 is 3 marks)

- 12** A satellite dish is fixed to a support bracket mounted on a flat rooftop. The support bracket AB is vertical and the strut BC holds the dish in place, as shown in the diagram.

$AB = 4.2$ m and $AC = 6.1$ m, where angle A is a right angle.



Show that the angle of elevation, x° , of the strut BC from the rooftop is 34.5° , correct to 1 decimal place.

(3)

(Total for Question 12 is 3 marks)

- 13** Two mathematically similar water containers are used to mix paint for a Model United Nations art installation designed by the school's UI/UX and Digital Art society.

The volume of the smaller container is $(3x + 6) \text{ cm}^3$.

The volume of the larger container is $(24x + 3) \text{ cm}^3$.

The ratio of the volume of the smaller container to the volume of the larger container is 1 : 9.

Given that the two containers are similar, find the value of x .

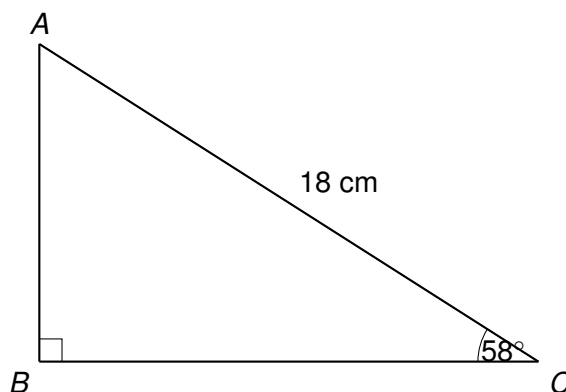
$$x = \dots\dots\dots$$

(4)

(Total for Question 13 is 4 marks)

- 14** A company designs a digital-art scanning stand used to photograph miniature models for a UI/UX product catalogue. Before manufacturing the full-size stand, the design team builds a small demonstration model that is mathematically similar to the full-size stand.

On the demonstration model, the support arm AC leans against the vertical column AB , touching the base at C . The support arm makes an angle of 58° with the base BC , and $AC = 18$ cm.



The full-size stand is a mathematically similar (enlarged) version of the demonstration model. The base BC of the full-size stand measures 220 cm, whereas the base BC of the demonstration model measures 66 cm.

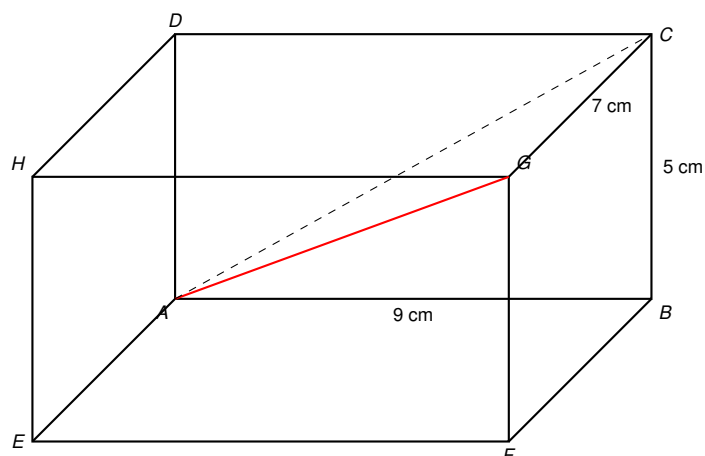
Work out the height AB of the vertical column on the **full-size** stand.

..... cm

(4)

(Total for Question 14 is 4 marks)

- 15** A school's Design Technology club is building a rectangular laboratory frame, $ABCDEFGH$, for a UI/UX display stand. The base $ABCD$ is a rectangle with $AB = 9$ cm and $BC = 5$ cm. The frame has a vertical height $CG = 7$ cm, as shown in the diagram.



A diagonal support rod is fixed from corner A (on the base) to corner G (diagonally opposite, at the top of the frame), as shown in red.

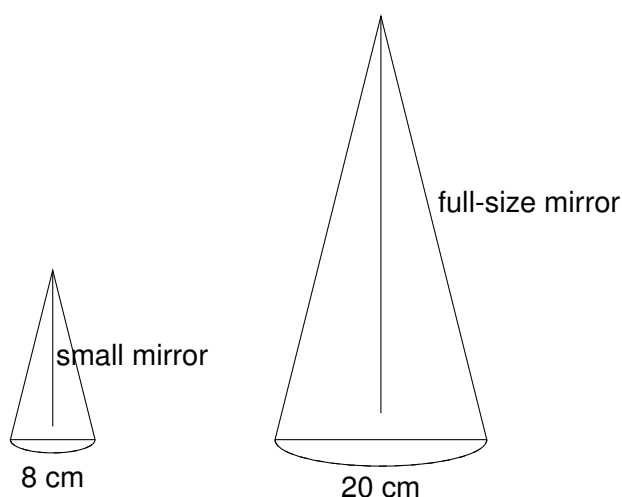
Find the angle that the rod AG makes with the base $ABCD$, giving your answer correct to 1 decimal place.

$$x = \dots\dots\dots^\circ$$

(4)

(Total for Question 15 is 4 marks)

- 16** A telescope manufacturer produces two similar cone-shaped mirror blanks, a small prototype and a full-size mirror, for testing before grinding into parabolic shape.



The small mirror blank has a base diameter of 8 cm. The full-size mirror blank has a base diameter of 20 cm. The two blanks are mathematically similar cones.

The curved surface area of the small mirror blank is 150.8 cm^2 . The volume of the full-size mirror blank is 2400 cm^3 .

- (a) Find the linear scale factor from the small mirror blank to the full-size mirror blank.

.....

(1)

- (b) Calculate the curved surface area of the full-size mirror blank. Give your answer to 3 significant figures.

..... cm^2

(2)

- (c) Calculate the volume of the small mirror blank. Give your answer to 3 significant figures.

..... cm³

(2)

(Total for Question 16 is 5 marks)

TOTAL FOR PAPER IS ?? MARKS